

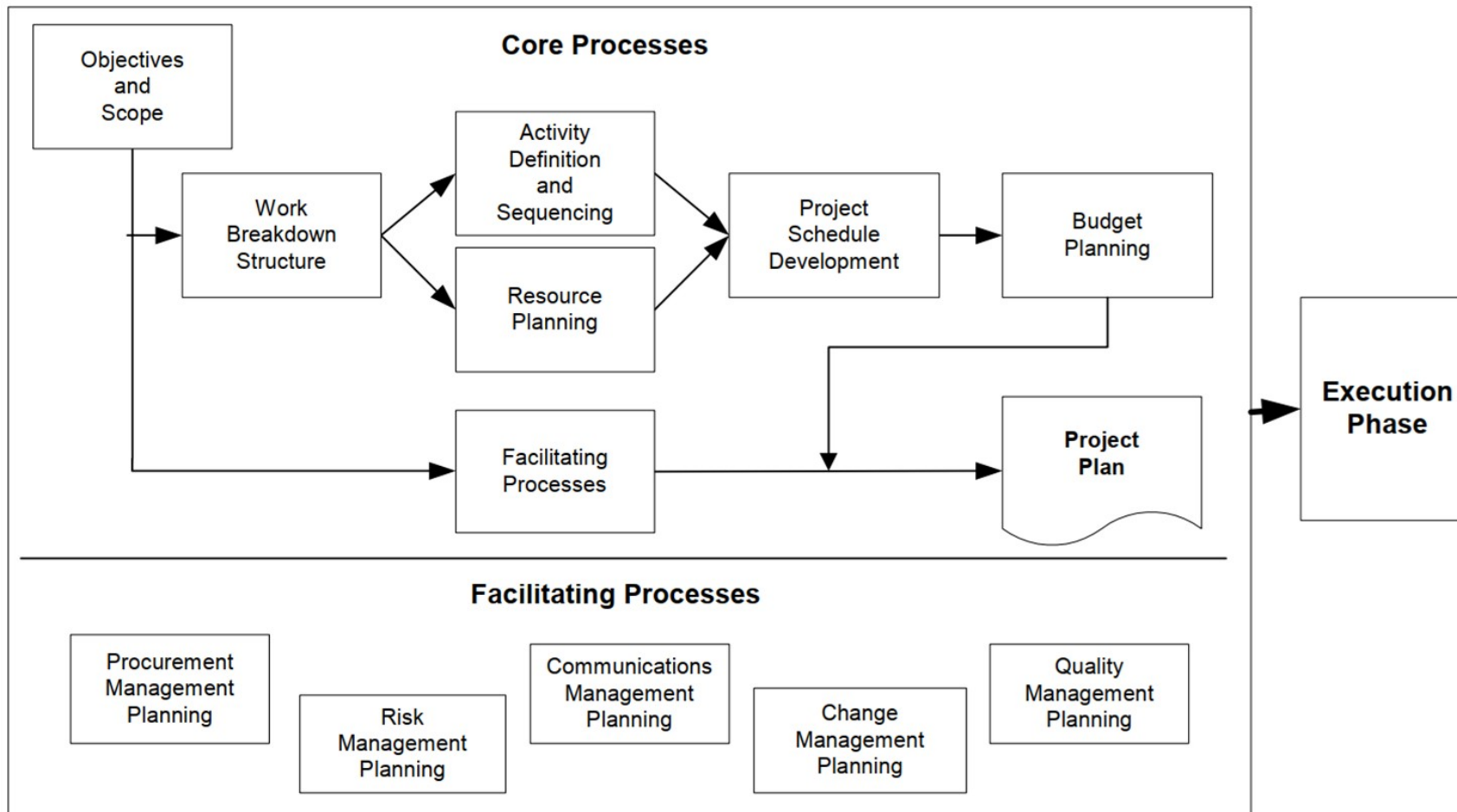
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# IT Project Management Support

Lecture 6

## Project Scheduling

## Project time management



## Project time management

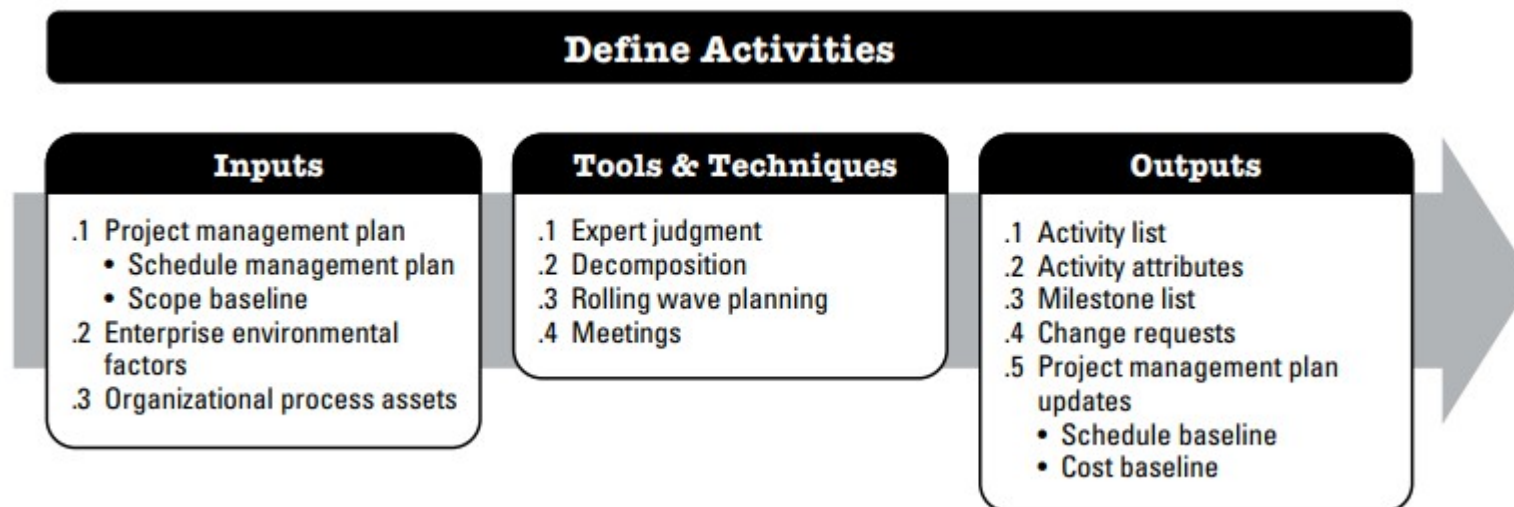
- Managers often cite the need to deliver projects on time as one of their biggest challenges and the main cause of conflict.
- Perhaps part of the reason that schedule problems are so common is that time is easily measured and remembered.
- Once a project schedule is set, people remember the projected completion date, and anyone can quickly estimate schedule performance by subtracting the original time estimate from how long it really took to complete the project.
- People often compare planned and actual project completion times without taking into account the approved changes in the project.
- Project time management, simply defined, involves the processes required to ensure timely completion of a project.
- Seven main processes are involved in project time management according to the PMBOK Guide:

1. **Planning schedule management** involves determining the policies, procedures, and documentation that will be used for planning, executing, and controlling the project schedule. The main output of this process is a schedule management plan.
2. **Defining activities** involves identifying the specific activities that the project team members and stakeholders must perform to produce the project deliverables. An activity or task is an element of work normally found on the work breakdown structure (WBS) that has expected duration, cost, and resource requirements. The main outputs of this process are an activity list, activity attributes, a milestone list, and project management plan updates.
3. **Sequencing activities** involves identifying and documenting the relationships between project activities. The main outputs of this process include project schedule network diagrams and project documents updates.

4. **Estimating activity resources** involves estimating how many resources - people, equipment, and materials - a project team should use to perform project activities. The main outputs of this process are activity resource requirements, a resource breakdown structure, and project documents updates.
5. **Estimating activity durations** involves estimating the number of work periods that are needed to complete individual activities. Outputs include activity duration estimates and project documents updates.
6. **Developing the schedule** involves analyzing activity sequences, activity resource estimates, and activity duration estimates to create the project schedule. Outputs include a schedule baseline, project schedule, schedule data, project calendars, project management plan updates, and project documents updates.
7. **Controlling the schedule** involves controlling and managing changes to the project schedule. Outputs include work performance information, schedule forecasts, change requests, project management plan updates, project documents updates, and organizational process assets updates.

## Defining activities

- Define Activities is the process of identifying and documenting the specific actions to be performed to produce the project deliverables.
- The key benefit of this process is that it decomposes work packages into schedule activities that provide a basis for estimating, scheduling, executing, monitoring, and controlling the project work.



## Defining activities

- You might think that all project work has been defined in enough detail after planning scope management, but it is often necessary to describe activities in more detail as part of schedule management.
- Defining activities involves identifying the specific actions that will produce the project deliverables in enough detail to determine resource and schedule estimates.
- The project team reviews the schedule management plan, scope baseline which includes the WBS, enterprise environmental factors, and organizational process assets to begin defining activities.
- Outputs of this process include an [activity list](#), [activity attributes](#), [a milestone list](#), and project management plan updates.

## Defining activities

- **Activity list** - is a tabulation of activities to be included on a project schedule. The list should include the activity name, an activity identifier or number, and a brief description of the activity.
- **Activity attributes** - provide schedule-related information about each activity, such as predecessors, successors, logical relationships, leads and lags, resource requirements, constraints, imposed dates, and assumptions related to the activity.
- The activity list and activity attributes should agree with the WBS and WBS dictionary.
- Information is added to the activity attributes as it becomes available; this information includes logical relationships and resource requirements that are determined in later processes.
- Many project teams use an automated system to keep track of activity-related information, e.g. [GanttPRO](#), [Ms Project](#), and more.



## Defining activities

- **Milestone** on a project - is a significant event that normally has no duration.
- It often takes several activities and a lot of work to complete a milestone, but the milestone itself is a marker to help in identifying necessary activities.
- Milestones are also useful tools for setting schedule goals and monitoring progress.
- For example, milestones on a project might include:
  - completion and customer sign-off of documents, such as design documents and test plans;
  - completion of specific products, such as software modules or installation of new hardware;
  - and completion of important process-related work, such as project review meetings and tests.
- Milestones are the most important and visible events.

## Defining activities

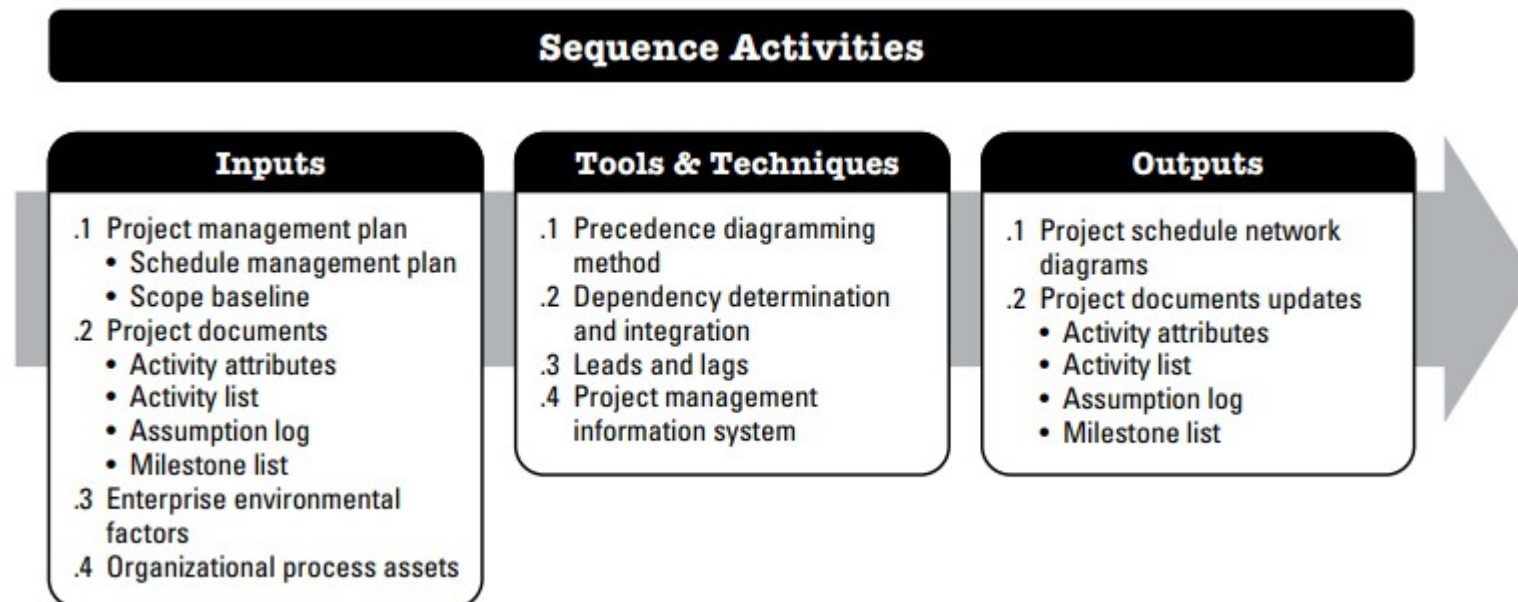
- The goal of defining activities is to ensure that the project team completely understands all the work it must do as part of the project scope so they can start scheduling the work.
- For example, a WBS item might be “Produce study report.”
- The project team must understand what the item means before team members can make schedule-related decisions.
  - How long should the report be?
  - Does it require a survey or extensive research to produce?
  - What skill level does the report writer need to have?
- Further defining the task will help the project team determine how long it will take to do and who should do it.

## Defining activities

- The WBS is often dissected further as the project team members continue to define the activities required for performing the work.
- For example, the task “Produce study report” might be broken down into several subtasks describing the steps of producing the report, such as
  - developing a survey,
  - administering the survey,
  - analyzing the survey results,
  - performing research,
  - writing a draft report, editing the report,
  - and finally producing the report.
- The project team should review the activity list and activity attributes with project stakeholders before moving on to the next step in project time management.
- If the team does not review these items with project stakeholders, it could produce an unrealistic schedule and deliver unacceptable results.

## Sequencing activities

- Sequencing Activities is the process of identifying and documenting relationships among the project activities.
- The key benefit of this process is that it defines the logical sequence of work to obtain the greatest efficiency given all project constraints.



## Sequencing activities

- After defining project activities, the next step in project time management is sequencing them or determining their dependencies.
- Inputs to the activity sequencing process include the schedule management plan, activity list and attributes, project scope statement, milestone list, and organizational process assets.
- The sequencing process involves evaluating
  - the reasons for dependencies and
  - the different types of dependencies.

## Dependencies

- A dependency or relationship pertains to the sequencing of project activities or tasks.
- For example, does a certain activity have to be finished before another can start?
- Can the project team do several activities in parallel? Can some overlap?
- Determining these relationships or dependencies among activities is crucial for developing and managing a project schedule.
- There are three basic reasons for identifying dependencies among project activities:
  1. Mandatory dependencies
  2. Discretionary dependencies
  3. External dependencies

## 1. Mandatory dependencies

- Mandatory dependencies are inherent in the nature of the work being performed on a project.
- They are sometimes referred to as hard logic.
- For example, you cannot test code until after the code is written.

## 2. Discretionary dependencies

- Discretionary dependencies are defined by the project team.
- For example, a project team might follow good practice and not start the detailed design of a new information system until the users sign off on all of the analysis work.
- Discretionary dependencies are sometimes referred to as soft logic and should be used with care because they may limit later scheduling options.



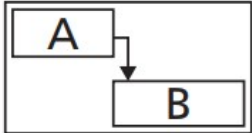
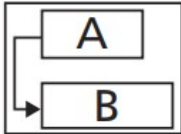
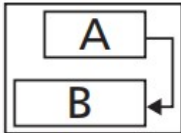
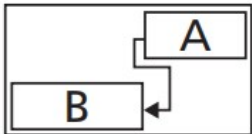
### 3. External dependencies

- External dependencies involve relationships between project and non-project activities.
- For example, the installation of a new operating system and other software may depend on delivery of new hardware from an external supplier.
- Even though delivery of the hardware may not be included in the scope of the project, you should add an external dependency to it because late delivery will affect the project schedule.

## Precedence Diagramming Method (PDM)

- The Precedence Diagramming Method (PDM) is a technique used for constructing a schedule model in which activities are represented by nodes and are graphically linked by one or more logical relationships to show the sequence in which the activities are to be performed.
- PDM includes four types of dependencies or logical relationships.
- A predecessor activity is an activity that logically comes before a dependent activity in a schedule.
- A successor activity is a dependent activity that logically comes after another activity in a schedule.
- These relationships are defined and illustrated below.

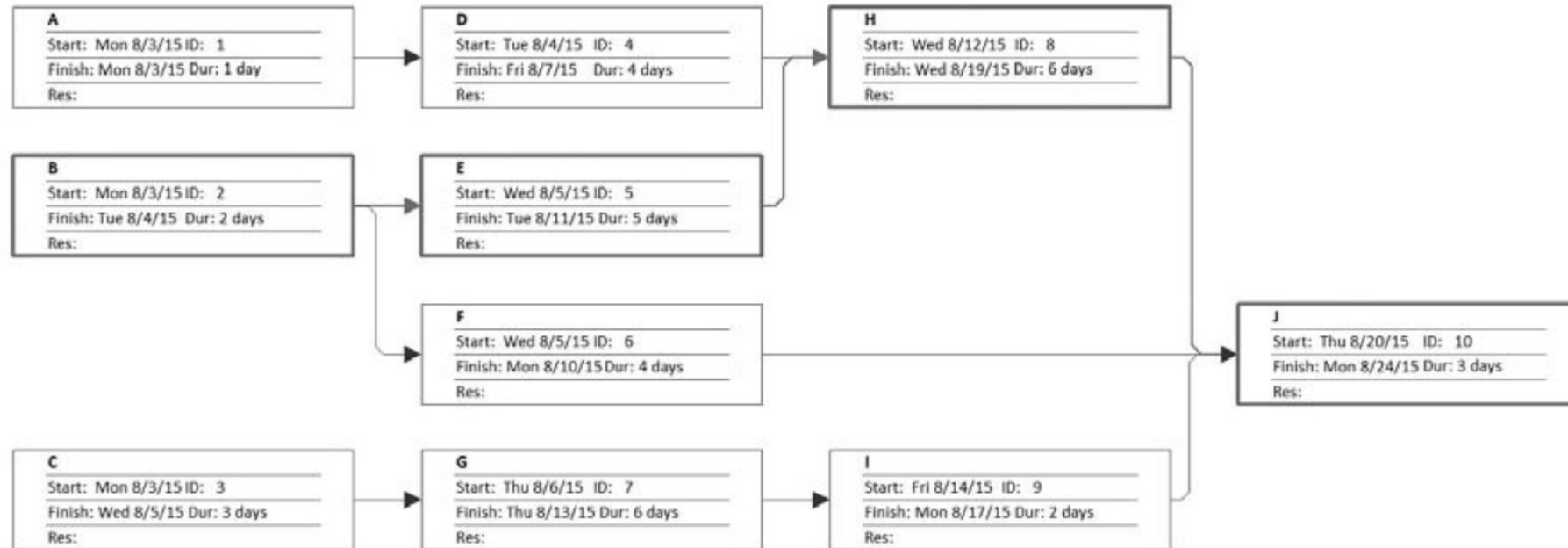
## Precedence Diagramming Method (PDM)

| Task dependency       | Example                                                                              | Description                                     |
|-----------------------|--------------------------------------------------------------------------------------|-------------------------------------------------|
| Finish-to-start (FS)  |    | Task (B) cannot start until task (A) finishes.  |
| Start-to-start (SS)   |    | Task (B) cannot start until task (A) starts.    |
| Finish-to-finish (FF) |   | Task (B) cannot finish until task (A) finishes. |
| Start-to-finish (SF)  |  | Task (B) cannot finish until task (A) starts.   |

## Precedence Diagramming Method (PDM)

- **Finish-to-start dependency:** A relationship in which the “from” activity or predecessor must finish before the “to” activity or successor can start. For example, you cannot provide user training until after software or a new system has been installed. Finish-to-start is the most common type of relationship or dependency.
- **Start-to-start dependency:** A relationship in which the “to” activity cannot start until the “from” activity or predecessor is started. For example, on IT projects, a group of activities might start simultaneously, such as the many tasks that occur when a new system goes live.
- **Finish-to-finish dependency:** A relationship in which the “from” activity must be finished before the “to” activity can be finished. One task cannot finish before another finishes. For example, quality control efforts cannot finish before production finishes, although the two activities can be performed at the same time.
- **Start-to-finish dependency:** A relationship in which the “from” activity must start before the “to” activity can be finished. This type of relationship is rarely used, but it is appropriate in some cases.

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Example of precedence diagramming method (PDM) network diagram for Project X

- Activities are placed inside boxes, which represent the nodes on this diagram.
- Arrows show the relationships between activities.
- Each task box includes the start and finish dates, which are labeled Start and Finish; the task ID number, labeled ID; the task's duration, labeled Dur; and the names of resources, if any, that are assigned to the task. These resources are labeled Res.

## Estimating activity resources (1)

- Before you can estimate the duration for each activity, you must have a good idea of the quantity and type of resources (people, equipment, and materials) that will be assigned to each activity.
- The nature of the project and the organization will affect resource estimates.
- Expert judgment, an analysis of alternatives, estimating data, and project management software are tools that can assist in resource estimating.
- The people who help determine what resources are necessary must have experience and expertise in similar projects and with the organization performing the project.

## Estimating activity resources (2)

Important questions to answer when estimating activity resources include:

- How difficult will specific activities be on this project?
- Is anything unique in the project's scope statement that will affect resources?
- What is the organization's history in doing similar activities? Has the organization done similar tasks before? What level of personnel did the work?
- Does the organization have people, equipment, and materials that are capable and available for performing the work? Could any organizational policies affect the availability of resources?
- Does the organization need to acquire more resources to accomplish the work? Would it make sense to outsource some of the work? Will outsourcing increase or decrease the amount of resources needed and when they will be available?

## Estimating activity resources (3)

- Answering these questions requires important inputs such as a project's schedule management plan, activity list, activity attributes, resource calendars, risk register, activity cost estimates, enterprise environmental factors, and organizational process assets such as policies regarding staffing and outsourcing.
- During the early phases of a project, the project team may not know which specific people, equipment, and materials will be available.
- For example, the team might know from past projects that a mix of experienced and inexperienced programmers will work on a project.
- The team might also be able to approximate the number of people or hours needed to perform specific activities.



## Estimating activity resources (4)

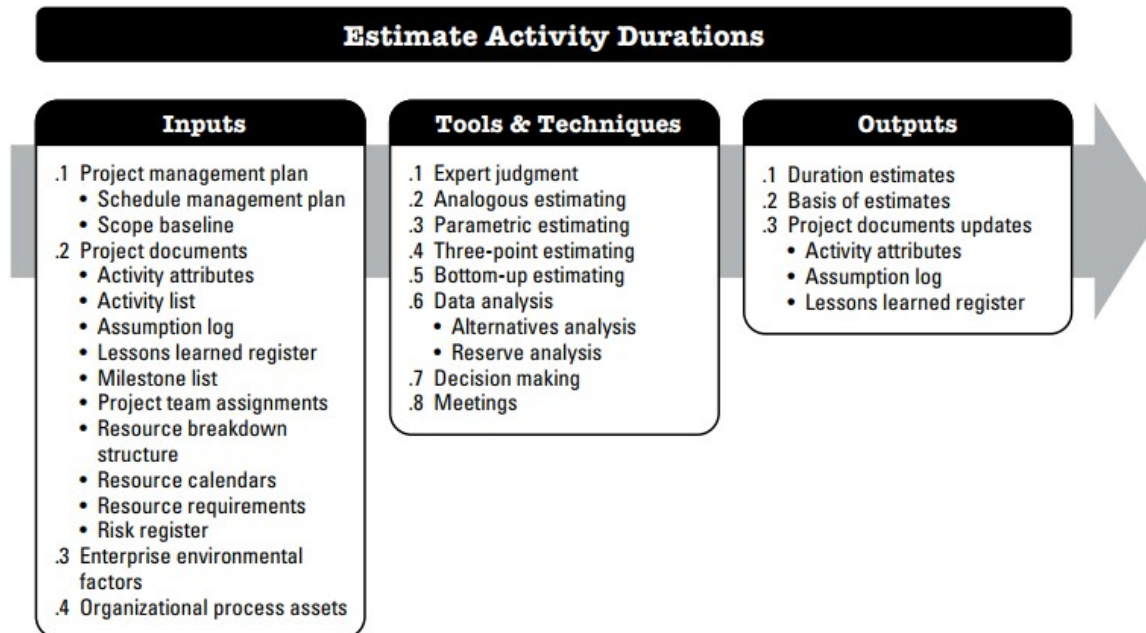
- It is important to thoroughly brainstorm and evaluate alternatives related to resources, especially on projects that involve people from multiple disciplines and companies.
- Because most projects involve many human resources and the majority of costs are for salaries and benefits, it is often effective to solicit ideas from different people to help develop alternatives and address resource-related issues early in a project.
- The resource estimates should also be updated as more detailed information becomes available.

## Estimating activity resources (5)

- The main outputs of the resource estimating process include a list of activity resource requirements, a resource breakdown structure, and project documents updates,
- A resource breakdown structure is a hierarchical structure that identifies the project's resources by category and type. Resource categories might include analysts, programmers, and testers.
- This information would be helpful in determining resource costs, acquiring resources, and so on.

## Estimating activity durations

- Estimating Activity Durations is the process of estimating the number of work periods needed to complete individual activities with estimated resources.
- The key benefit of this process is that it provides the amount of time each activity will take to complete.



## Estimating activity durations

- After working with key stakeholders to define activities, determine their dependencies, and estimate their resources, the next process in project time management is to estimate the duration of activities.
- It is important to note that duration includes the actual amount of time worked on an activity plus elapsed time.
- For example, even though it might take one workweek or five workdays to do the actual work, the duration estimate might be two weeks to allow extra time needed to obtain outside information.
- The people or resources assigned to a task will also affect the task duration estimate.
- As another example, if someone asked for an estimate of when you plan to finish reading a particular book, you might give an answer of two months.
- Two months would be the duration estimate, even if you only plan to spend 20 hours actually reading the book.

## Estimating activity durations

- There are several inputs to activity duration estimates, including
  - the schedule management plan,
  - activity list,
  - activity attributes,
  - lessons learned register
  - activity resource requirements,
  - resource calendars,
  - project scope statement,
  - risk register,
  - resource breakdown structure,
  - enterprise environmental factors,
  - and organizational process assets.
- In addition to reviewing past project information, the team should review the accuracy of the duration estimates thus far on the project.
- For example, if team members find that all of their estimates have been much too long or short, they should update the estimates to reflect what they have learned.

## Estimating activity durations

- One of the most important considerations in making activity duration estimates is the availability of resources, especially human resources.
- What specific skills do people need to do the work?
- What are the skill levels of the people assigned to the project?
- How many people are expected to be available to work on the project at any one time?

## Estimating activity durations – Tools and techniques

- Expert judgement.
- Analogous estimating.
- Three-point estimating.

## Expert judgement

Expertise should be considered from individuals or groups with specialized knowledge or training in the following topics:

- Schedule development, management, and control;
- Expertise in estimating; and
- Discipline or application knowledge.



## Analogous estimating (1)

- Analogous estimating is a technique for estimating the duration or cost of an activity or a project using historical data from a similar activity or project.
- Analogous estimating uses parameters from a previous, similar project, such as duration, budget, size, weight, and complexity, as the basis for estimating the same parameter or measure for a future project.
- When estimating durations, this technique relies on the actual duration of previous, similar projects as the basis for estimating the duration of the current project.
- It is a gross value estimating approach, sometimes adjusted for known differences in project complexity.

## Analogous estimating (2)

- Analogous duration estimating is frequently used to estimate project duration when there is a limited amount of detailed information about the project.
- Analogous estimating is generally less costly and less time-consuming than other techniques, but it is also less accurate.
- Analogous duration estimates can be applied to a total project or to segments of a project and may be used in conjunction with other estimating methods.
- Analogous estimating is most reliable when the previous activities are similar and the project team members preparing the estimates have the needed expertise.

## Three-point estimating

- The accuracy of single-point duration estimates may be improved by considering estimation uncertainty and risk.
- Using three-point estimates helps define an approximate range for an activity's duration:
- **Most likely ()**. This estimate is based on the duration of the activity, given the resources likely to be assigned, their productivity, realistic expectations of availability for the activity, dependencies on other participants, and interruptions.
- **Optimistic ()**. The activity duration based on analysis of the best-case scenario for the activity.
- **Pessimistic ()**. The duration based on analysis of the worst-case scenario for the activity.
- Depending on the assumed distribution of values within the range of the three estimates, the expected duration (), can be calculated.

## Three-point estimating

### Triangular distribution.

- One commonly used formula is triangular distribution (arithmetic average):

$$= ( o + m + p ) / 3$$

where o=optimistic, m=most likely, p=pessimistic

- Triangular distribution is used when there is insufficient historical data or when using judgmental data.
- Duration estimates based on three points with an assumed distribution provide an expected duration and clarify the range of uncertainty around the expected duration.

## Three-point estimating

### PERT method.

- When there is a high degree of uncertainty about the individual activity duration estimates, the network analysis Program Evaluation and Review Technique (PERT) can be used to estimate project duration.
- PERT uses probabilistic time estimates - duration estimates based on using **optimistic, most likely, and pessimistic estimates of activity durations** - instead of one specific or discrete duration estimate
- To use PERT, you calculate a weighted average for the duration estimate of each project activity using the following formula:

$$= ( o + 4m + p ) / 6$$

where o=optimistic, m=most likely, p=pessimistic

- By using the PERT weighted average for each activity duration estimate, the total project duration estimate takes into account the risk or uncertainty in the individual activity estimates.

## Bottom-up estimating

- Bottom-up estimating is a method of estimating project duration (or cost) by aggregating the estimates of the lower-level components of the WBS.
- When an activity's duration cannot be estimated with a reasonable degree of confidence, the work within the activity is decomposed into more detail.
- The detail durations are estimated.
- These estimates are then aggregated into a total quantity for each of the activity's durations.

## Data analysis

- Data analysis techniques that can be used for this process include but are not limited to:
- **Alternatives analysis.** Alternatives analysis is used to compare various levels of resource capability or skills; scheduling compression techniques ; different tools (manual versus automated); and make, rent, or buy decisions regarding the resources. This allows the team to weigh resource, cost, and duration variables to determine an optimal approach for accomplishing project work.
- **Reserve analysis.** Reserve analysis is used to determine the amount of contingency and management reserve needed for the project. Duration estimates may include contingency reserves, sometimes referred to as schedule reserves, to account for schedule uncertainty. Contingency reserves are the estimated duration within the schedule baseline, which is allocated for identified risks that are accepted

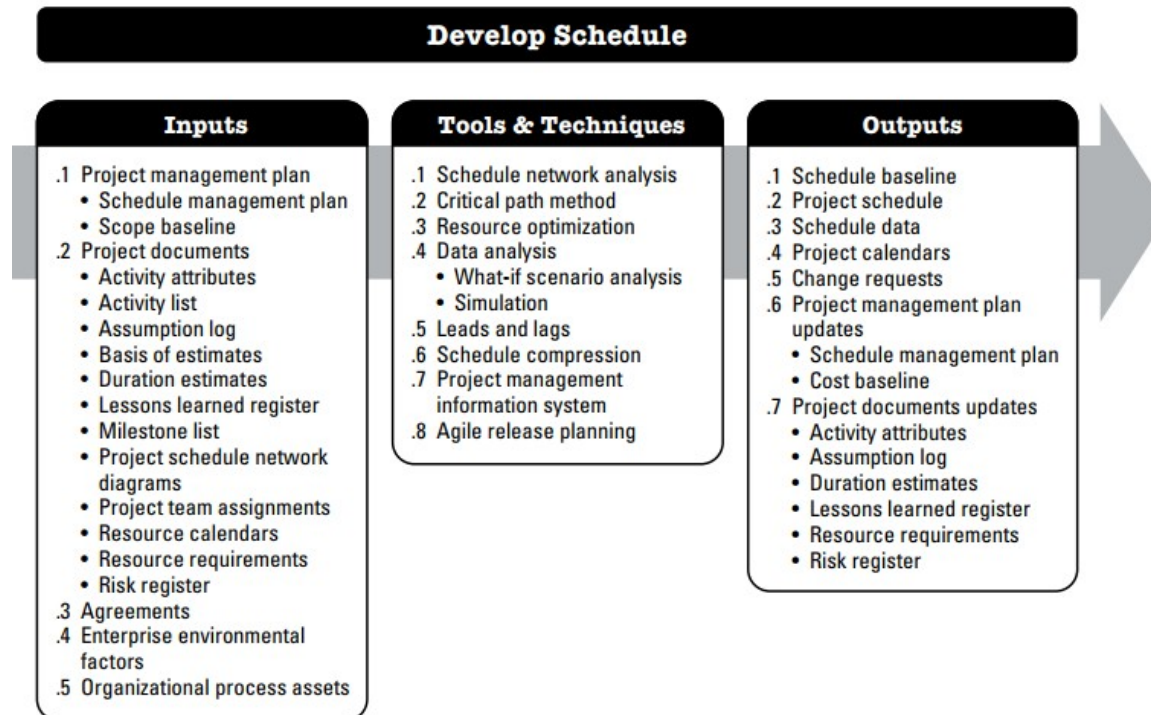
## Techniques used in agile-based projects

- Techniques used in agile-based projects will be covered in one of the next lectures:
- **Decision-making techniques**
- **Meetings**



## Developing schedule

- Developing schedule is the process of analyzing activity sequences, durations, resource requirements, and schedule constraints to create a schedule model for project execution and monitoring and controlling.
- The key benefit of this process is that it generates a schedule model with planned dates for completing project activities.



## Developing schedule

- Developing an acceptable project schedule is an iterative process.
- The schedule model is used to determine the planned start and finish dates for project activities and milestones based on the best available information.
- Schedule development can require the review and revision of duration estimates, resource estimates, and schedule reserves to establish an approved project schedule that can serve as a baseline to track progress.
- Key steps include defining the project milestones, identifying and sequencing activities, and estimating durations.

## Developing schedule

- Schedule development uses the results of all the preceding project time management processes to determine the start and end dates of the project and its activities.
- Project time management processes often go through several iterations before a project schedule is finalized.
- The ultimate goal of developing a realistic project schedule is to provide a basis for monitoring project progress for the time dimension of the project.
- The main outputs of this process are the project schedule, a schedule baseline, schedule data, project calendars, project management plan updates, and project documents updates.
- Some project teams create a computerized model to create a network diagram, enter resource requirements and availability by time period, and adjust other information to quickly generate alternative schedules.

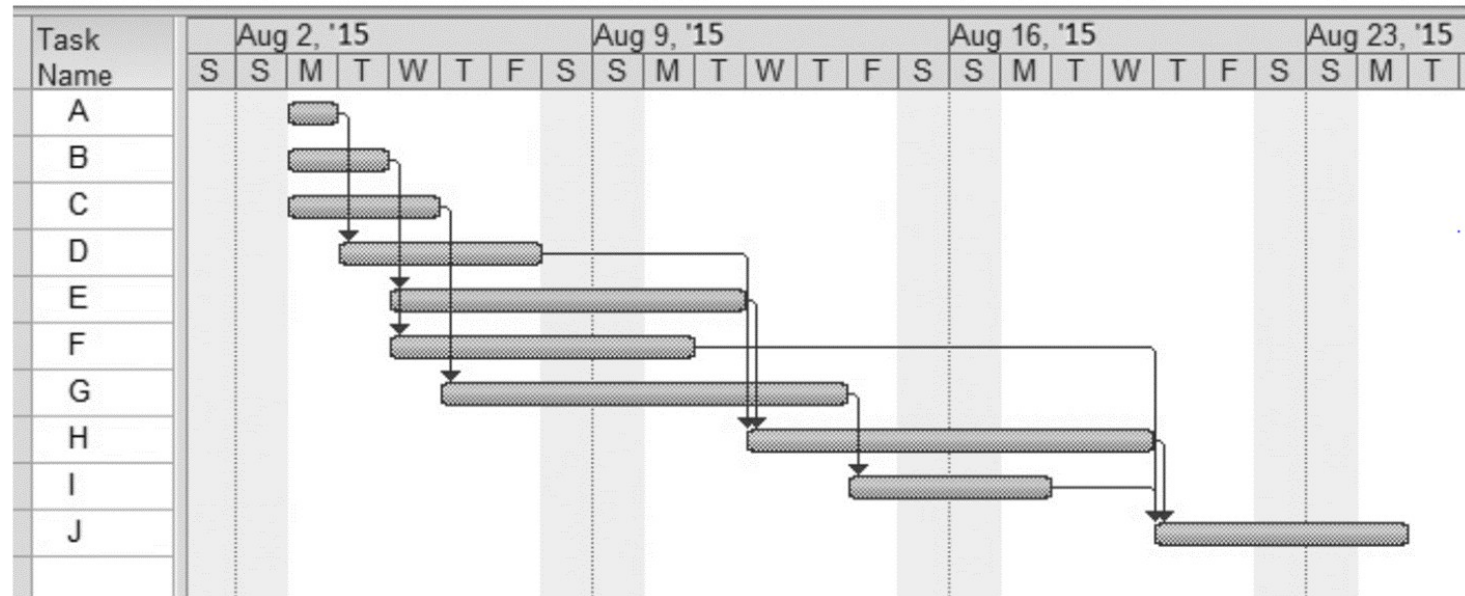
## Developing schedule

Several tools and techniques assist in schedule development:

- **Gantt chart** is a common tool for displaying project schedule information.
- **Critical path analysis** is a very important tool for developing and controlling project schedules.
- **Critical chain scheduling** is a technique that focuses on limited resources when creating a project schedule.
- **PERT analysis** is a means for considering schedule risk on projects.

## Gantt Chart

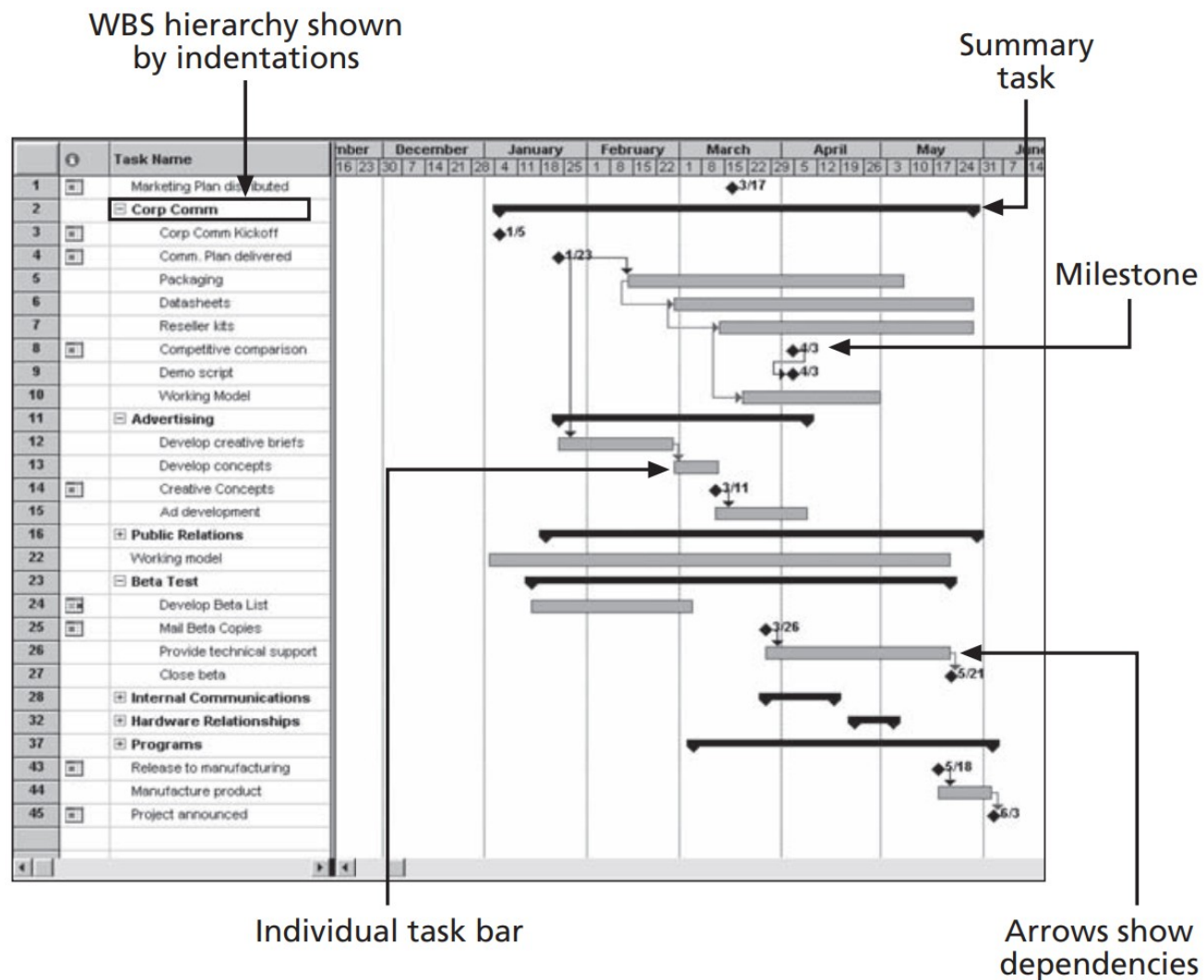
- Gantt charts provide a standard format for displaying project schedule information by listing project activities and their corresponding start and finish dates in calendar form.
- Gantt charts are sometimes referred to as bar charts because the activities' start and end dates are shown as horizontal bars.
- Gantt charts, represent schedule information where activities are listed on the vertical axis, dates are shown on the horizontal axis, and activity durations are shown as horizontal bars placed according to start and finish dates.
- Figure shows a simple Gantt chart for Project X created with Microsoft Project.



## Adding milestones to Gantt Chart

- Milestones can be a particularly important part of schedules, especially for large projects.
- Many people like to focus on meeting milestones, so you can create them to emphasize important events or accomplishments on projects.
- Normally, you create milestones by entering tasks with zero duration.
- The black diamond represents a milestone in a Gantt chart.
- To make milestones meaningful, some people use the SMART criteria to help define them. The SMART criteria are guidelines suggesting that milestones should be:
  - Specific
  - Measurable
  - Assignable
  - Realistic
  - Time-framed

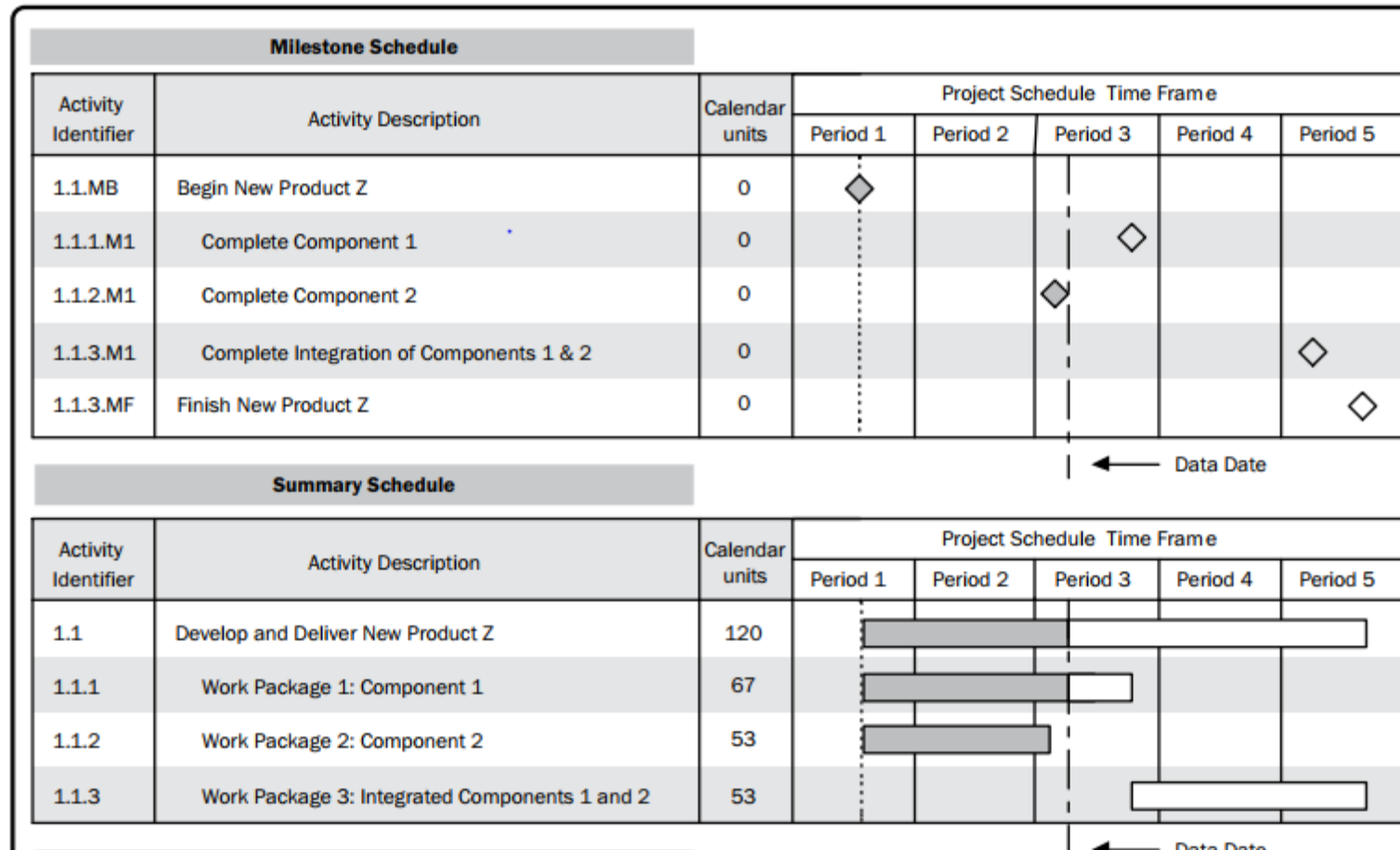
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## Gantt Chart

- For control and management communications, the broader, more comprehensive summary activity is used between milestones or across multiple interdependent work packages and is displayed in bar chart reports.



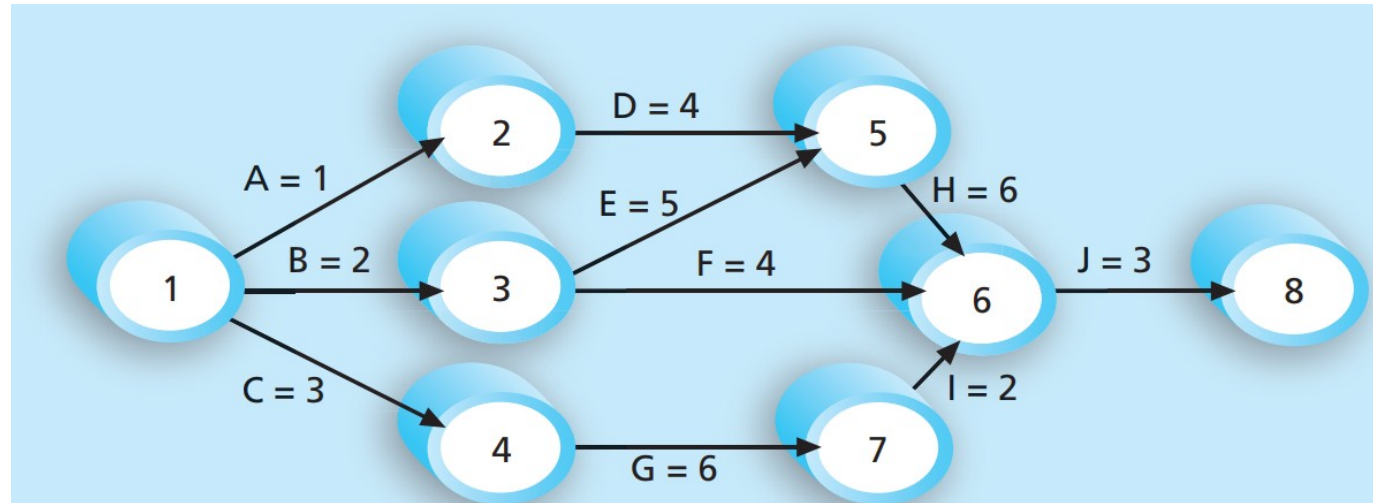


## Critical path method

- Critical path method (CPM) also called critical path analysis is a network diagramming technique used to predict total project duration. This important tool helps you combat project schedule overruns.
- A critical path for a project is the series of activities that determine the earliest time by which the project can be completed. It is the longest path through the network diagram and has the least amount of slack or float.
- Slack or float is the amount of time an activity may be delayed without delaying a succeeding activity or the project finish date.
- Normally, several tasks are done in parallel on projects, and most projects have multiple paths through a network diagram.
- The longest path or the path that contains the critical tasks is what drives the completion date for the project.
- You are not finished with the project until you have finished all the tasks.

## Critical path method (*AOA network diagram*)

- What does the critical path really mean? Even though the critical path is the longest path, it represents the shortest time required to complete a project.



Note: Assume all durations are in days.

Path 1: A-D-H-J Length =  $1+4+6+3 = 14$  days

Path 2: B-E-H-J Length =  $2+5+6+3 = 16$  days

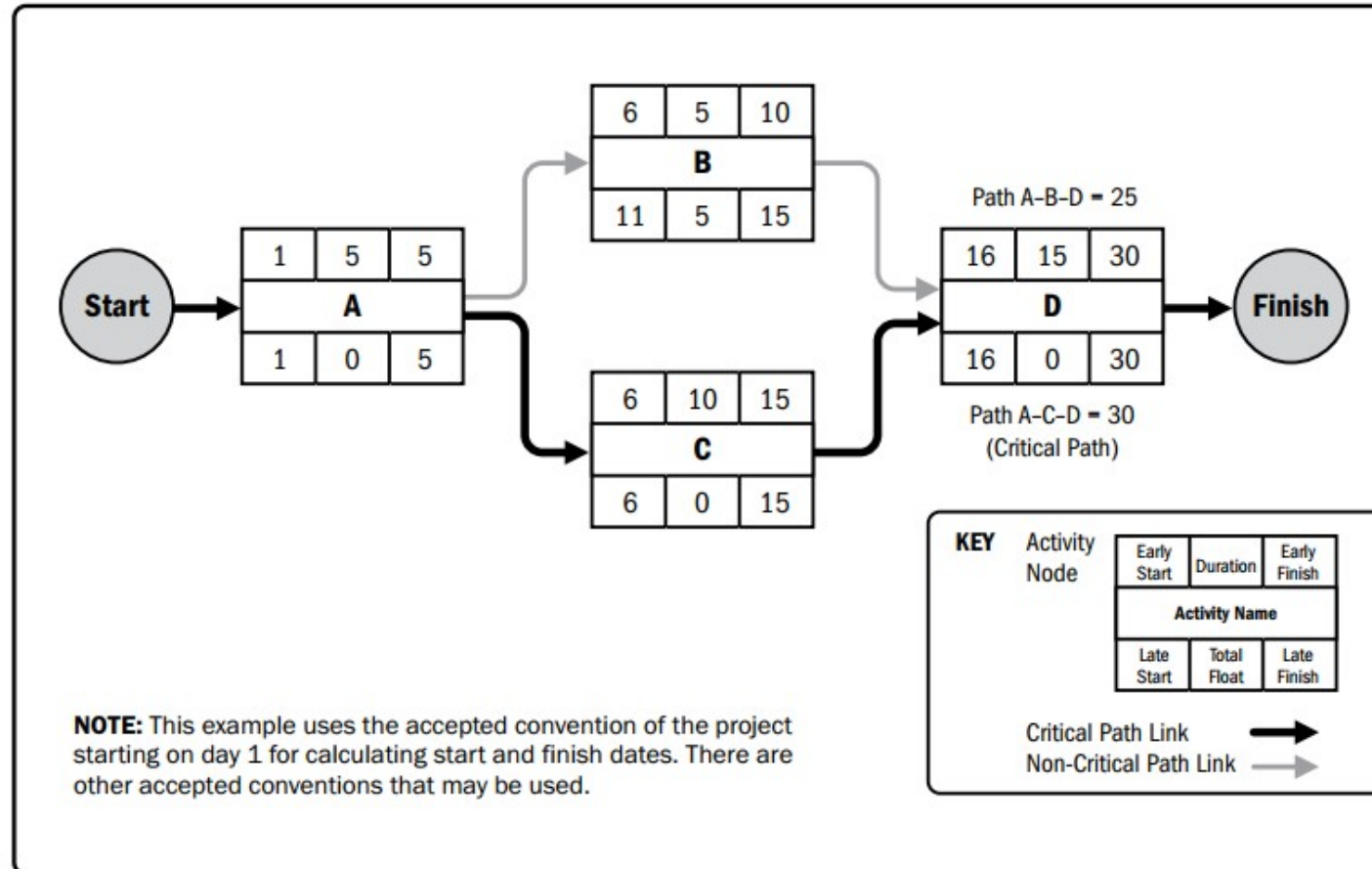
Path 3: B-F-J Length =  $2+4+3 = 9$  days

Path 4: C-G-I-J Length =  $3+6+2+3 = 14$  days

Since the critical path is the longest path through the network diagram, Path 2, B-E-H-J, is the critical path for Project X.

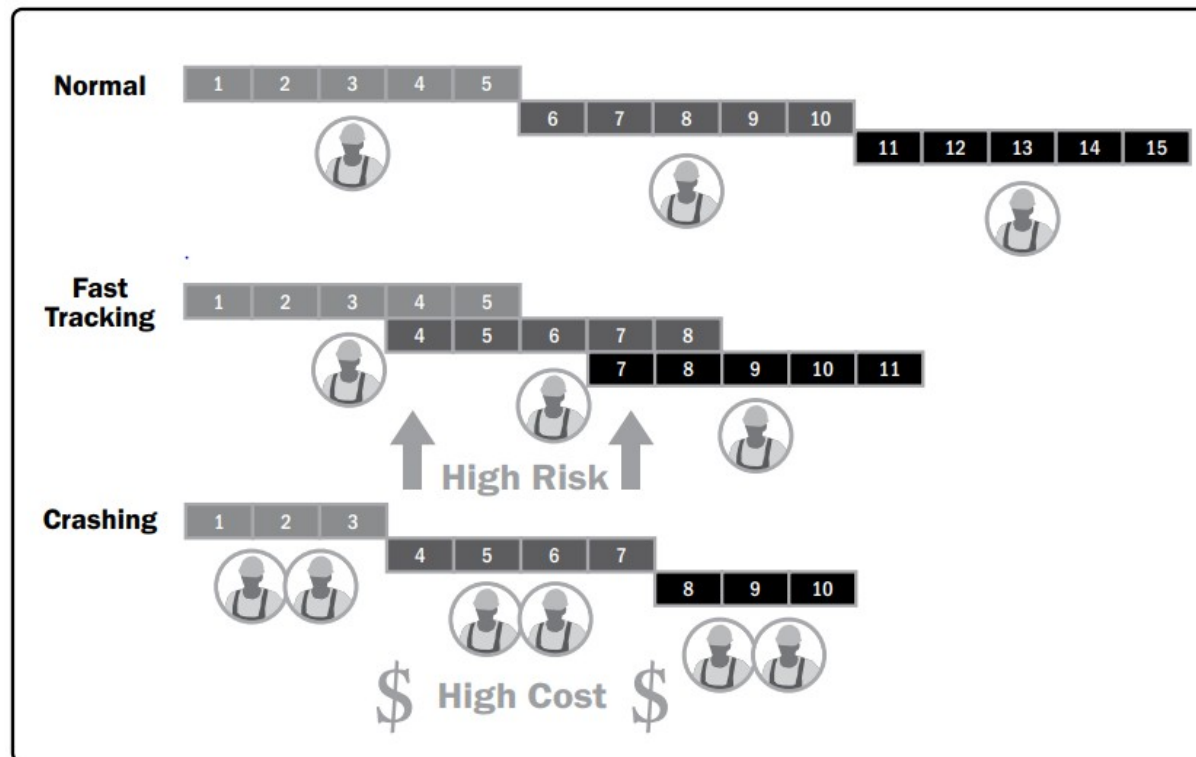
## Critical path method (*PDM network diagram*)

- Below another example of critical path method.



## Schedule compression

- Schedule compression techniques are used to shorten or accelerate the schedule duration without reducing the project scope in order to meet schedule constraints, imposed dates, or other schedule objectives.
- Schedule compression techniques include: **Crashing** and **Fast Tracking**.



## Schedule compression

### Crashing

- A technique used to shorten the schedule duration for the least incremental cost by adding resources. Examples of crashing include approving overtime, bringing in additional resources, or paying to expedite delivery to activities on the critical path. Crashing works only for activities on the critical path where additional resources will shorten the activity's duration. Crashing does not always produce a viable alternative and may result in increased risk and/or cost.

### Fast Tracking

- A schedule compression technique in which activities or phases normally done in sequence are performed in parallel for at least a portion of their duration. Fast tracking may result in rework and increased risk. Fast tracking only works when activities can be overlapped to shorten the project duration on the critical path. Fast tracking may also increase project costs.



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Thank You very much  
for your attention